

BSc CSIT – Object-Oriented Programming in C++

Practical / Laboratory Questions

Lab 1: Basic C++ Programming

1. Write a program to find the largest among three numbers using conditional statements.
2. Write a program to find the factorial of a number using a loop.
3. Write a program to calculate the sum of digits of a number.

Lab 2: Functions

4. Write a C++ program to find the greatest of two numbers using a function.
5. Write a program to demonstrate call by value and call by reference.
6. Write a program to find the maximum element of an array using a function.
7. Write a program demonstrating function overloading.

Lab 3: Classes and Objects

8. Create a class Employee with employee ID, name, and salary. Display employee details using member functions.
9. Write a program to demonstrate the difference between public, private, and protected members.

Lab 4: Constructors and Destructors

10. Write a program to demonstrate a default constructor.
11. Write a program to demonstrate a parameterized constructor.
12. Write a program demonstrating a copy constructor.
13. Create a class Student having multiple constructors and demonstrate constructor overloading.

Lab 5: Inheritance

14. Write a program demonstrating single inheritance using Person and Student.
15. Write a program demonstrating multilevel inheritance using Person → Employee → Manager.

Lab 6: Polymorphism and Function Overloading

16. Write a program to demonstrate function overloading.
17. Write a program to demonstrate operator overloading.
18. Write a program demonstrating compile-time polymorphism.

Lab 7: Virtual Functions and Runtime Polymorphism

19. Write a program to demonstrate the difference between a normal function and a virtual function.
20. Write a program using a pure virtual function.

Lab 8: Friend Function and Static Members

21. Write a program to demonstrate a friend function.
22. Write a program demonstrating a static member function.

Lab 9: Pointers and Dynamic Memory

23. Write a program to demonstrate pointers with objects.
24. Dynamically allocate memory for an integer using new and release it using delete.

Lab 10: Templates

25. Write a function template to find the greatest of two numbers.
26. Create a class template for performing arithmetic operations on two numbers.

Lab 11: Exception Handling

27. Write a program demonstrating try, throw, and catch.

Lab 12: File Handling

28. Write a C++ program to create and write data into a text file.
29. Write a program to read data from a file and display it.

Note: Do the above program in the system and write it in A4 paper with input and output. It is recommended to take the screenshot of each program and paste it. In each program write the outputting screen shot with simple cut statement saying "Author by **Your Name**"

Important Practical Questions

1. Class and object – Student/Employee
2. Constructor and destructor
3. Function overloading
4. Operator overloading
5. Single and multilevel inheritance
6. Multiple inheritance
7. Virtual function and runtime polymorphism
8. Pure virtual function / abstract class
9. Friend function
10. Static data member
11. Pointer to object and this pointer

12. Function template
13. Class template
14. Exception handling
15. File handling
16. Integrated Student/Bank/Library Management System